20 · Case 15: cystic fibrosis, Huntington's, and the two whānau

The words of this track, as it says them. Bold brown is the examiner.

Exam case 15. Genetic diseases.

You're working in a rural GP clinic, alongside two families with genetic diseases. The Harris whānau has cystic fibrosis. The Anderson whānau has Huntington's disease. You want to understand how these diseases pass from person to person, and how they might affect any future children born into these families.

One answered question, and four more on her two whānau charts, whose model answers are still hidden on Canvas. So you'll get the method for those: whatever the chart looks like, you'll be able to work it.

Six marks. What are the modes of inheritance of Huntington's disease and cystic fibrosis? Explain both: the chromosome type, the pattern of inheritance, and what this means.

Cystic fibrosis: autosomal recessive, on chromosome seven. It skips generations, and people can be carriers. The affected person is little r little r: homozygous recessive.

Huntington's disease: autosomal dominant, on chromosome four. It's expressed whenever the dominant allele is present, so in every generation someone is expected to have it. The affected person is big R little r, or big R big R.

When she marked a student's answer to this in her session, she added one thing in red: the answer must name the affected person's genotype. So always finish with it. Little r little r for cystic fibrosis. Big R little r, or big R big R, for Huntington's.

Now her two charts. Her model answers aren't posted yet, so here's how to work them. For the Harris whānau's chart, cystic fibrosis, state the genotype and phenotype of every person.

Cystic fibrosis is recessive, so start with the shaded people. Every affected person is little r little r. Next, the parents of an affected child who aren't affected themselves: each of them must have given a little r, so they're carriers, big R little r. Then the unaffected children of two carriers: big R big R, or big R little r. You can't tell which from the chart, so write both. For the phenotype, write affected, or unaffected. To be precise, an unaffected carrier.

The two youngest sisters of the Harris whānau. What's the chance they pass this condition on to their children, and what circumstances would have to occur?

The circumstance first: a child can only be affected if both parents pass on a little r. So the sister has to carry a little r, and so does her partner. If a sister is a carrier and her partner is a carrier, it's the classic cross: twenty-five per cent affected, fifty per cent carriers, twenty-five per cent clear. If a sister is affected, little r little r, every child gets a little r from her, so with a

carrier partner it's a fifty per cent chance for each child. And with a big R big R partner, none of the children are affected, though they'll be carriers.

For the Anderson whānau's chart, Huntington's, state the genotype and phenotype of every person.

Huntington's is dominant, so it's the other way round. On her chart the letters may be R and little r, or H and little h; the logic is the same. Every affected person is big H little h, or rarely big H big H. Every unaffected person is little h little h. There are no healthy carriers. And every affected person should have an affected parent.

The six cousins in the youngest generation of the Anderson whānau. What's the chance they could pass it on to their children?

Go cousin by cousin. An unaffected cousin is little h little h, so the chance is zero: they have no big H to pass on. An affected cousin is big H little h, and with an unaffected partner, it's a fifty per cent chance for each child. One real-world catch: Huntington's usually shows in adulthood, so a young cousin might not show it yet. But on a chart, go by the shading.

Once more. Case 15.

Cystic fibrosis is autosomal recessive, on chromosome seven: it skips generations, there are carriers, and the affected person is little r little r. Huntington's is autosomal dominant, on chromosome four: it shows in every generation, and the affected person is big R little r, or big R big R. On a recessive chart, label the affected people little r little r first; their unaffected parents are carriers. A child is affected only if both parents pass on a little r: two carriers give twenty-five per cent. On a dominant chart, affected is big H little h and unaffected is little h little h. An unaffected cousin can't pass it on; an affected one has a fifty per cent chance with each child.